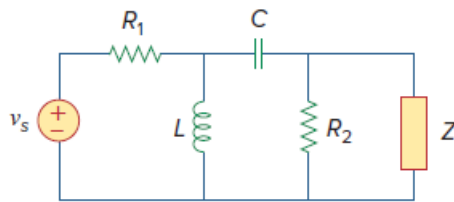


Problem

Using Fig. 11.45, design a problem to help other students better understand maximum average power transfer to a load Z .

Figure 11.45:



Step-by-step solution

Step 1 of 7

Consider the Figure 1 with the assumed value of R , L and C .

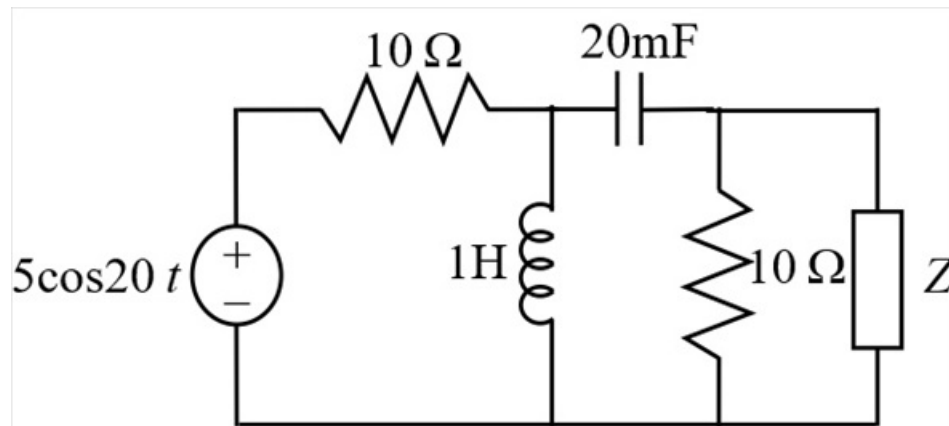


Figure 1

Step 2 of 7

To calculate the Thevenin impedance Z_{Th} , replace the voltage source by a short circuit as shown in Figure 2.

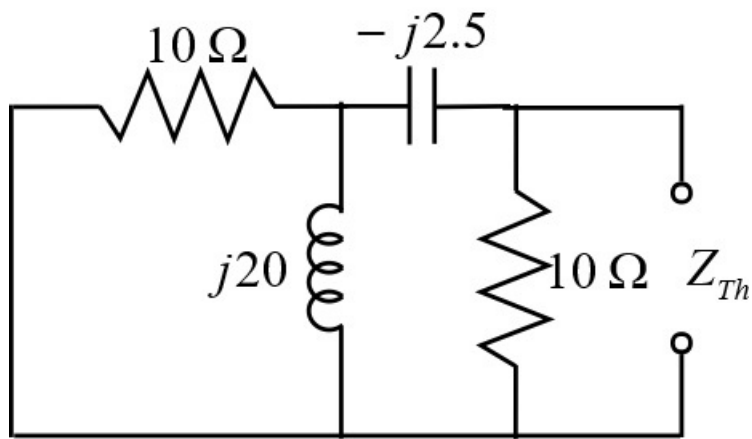


Figure 2

Step 3 of 7

Calculate the thevenin impedance Z_{Th} .

$$\begin{aligned} Z_{Th} &= [(10 \parallel j20) + (-j2.5)] \parallel 10 \\ &= \left[\frac{j200}{10 + j20} - j2.5 \right] \parallel 10 \\ &= \left[\frac{50 - j175}{10 + j20} \right] \parallel 10 \\ &= \left[\frac{500 - j175}{150 - j25} \right] \end{aligned}$$

Rewrite the above equation.

$$\begin{aligned} Z_{Th} &= 4.11 - j0.57 \\ &= 4.15 \angle -7.89^\circ \end{aligned}$$

Step 4 of 7

Calculate the thevenin voltage V_{Th} .

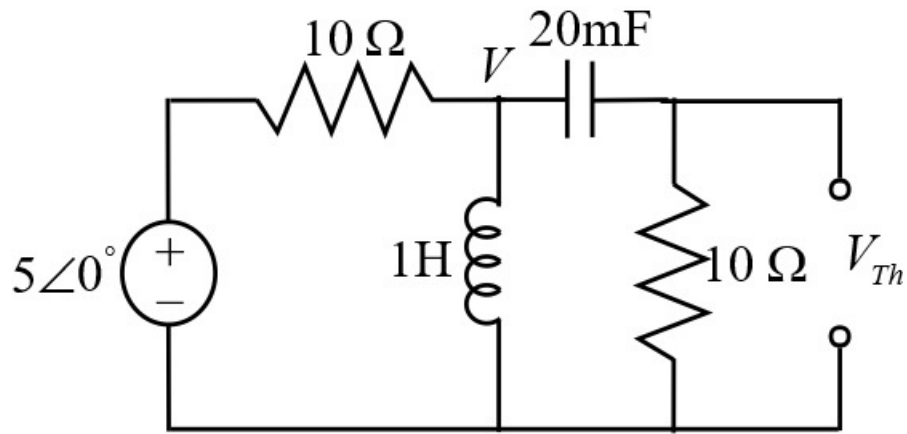


Figure 3

Step 5 of 7

Apply the nodal analysis at node V .

$$\frac{V-5}{10} + \frac{V}{20j} + \frac{V-V_{Th}}{-j2.5} = 0 \dots\dots (1)$$

Apply the nodal analysis at node V_{Th} .

$$\frac{V_{Th}-V}{-j2.5} + \frac{V_{Th}}{10} = 0$$

Rewrite the above equation.

$$V = [-2.5j + 10]V_{Th}$$

Substitute $[-2.5j + 10]V_{Th}$ for V in equation (1).

$$\begin{aligned} V_{Th} &= \frac{1}{6.45j + 5} \\ &= -.30 + 0.388j \end{aligned}$$

Step 6 of 7

Rewrite the above equation.

$$V_{Th} = 0.49\angle 127.7^\circ$$

Calculate the rms value of the voltage.

$$\begin{aligned} V_{rms} &= \frac{V_{Th}}{\sqrt{2}} \\ &= 0.346\angle 127.7^\circ \end{aligned}$$

Calculate the power factor.

$$\begin{aligned} pf &= \cos(-7.89) \\ &= 0.99 \end{aligned}$$

Step 7 of 7

Calculate the rms value of current.

$$\begin{aligned} I_{rms} &= \frac{V_{rms}}{Z} \\ &= \frac{0.346 \angle 127.7^\circ}{4.15 \angle -7.89^\circ} \\ &= 0.08 \angle 135.59^\circ \end{aligned}$$

The maximum average power to the load is,

$$\begin{aligned} P &= V_{rms} I_{rms} \text{pf} \\ &= (0.346)(0.08)0.99 \\ &= 27 \text{ mW} \end{aligned}$$

Hence, the maximum power transferred to the load is 27 mW.